



**SECD2523**

**DataBase**

**Project: Proposal**

**Title: Car Equipment Shop Database System**

**The team for this project consists of the following members:**

- ❖ **Mohammed Kamal**
- ❖ **Dina Mohammed**
- ❖ **Dyako Abubakr**

## 1. Introduction

The automotive parts industry is fast-paced and requires accurate, real-time data to manage inventory efficiently. This proposal outlines the development of a **Database Management System (DBMS)** specifically designed for a car equipment shop specializing in **Kia** and **Hyundai** parts. The system will streamline inventory management, provide flexible search capabilities, and improve overall operations by offering comprehensive tracking and categorization of car parts.

The database will be built to support core functions such as:

- **Detailed Inventory Management** by part category, stock levels, and availability status.
  - **Advanced Search and Filter Capabilities** to easily locate parts based on car brand, model, year, part location, and description.
  - **Flexible Year Range Selection** allowing the user to specify parts compatible across a range of model years, essential for parts that fit multiple versions of a vehicle.
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## 2. Project Objectives

The primary goal is to develop an intuitive, scalable database system that:

- **Manages Inventory Efficiently:** Tracks all parts by category, brand, model, year, stock levels, and availability.
  - **Enhances Search and Filter Options:** Enables precise search criteria, ensuring users can find parts quickly using detailed filters.
  - **Improves Accuracy:** Ensures that parts availability is always up to date, preventing overstock or shortages through real-time updates.
  - **Offers Flexibility:** Allows the shop to manage parts with or without specific stock counts and provides flexibility in setting year ranges for compatible parts.
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### 3. Functional Requirements

The system will provide a complete solution with the following core functionalities:

#### 3.1 Inventory Management

The inventory will be managed through a **structured database** that stores and organizes parts with the following details:

- **Part Name:** The name of the car part (e.g., "Hood").
- **Serial Code:** A unique identifier for each part to ensure precise tracking.
- **Price:** The cost of the part.
- **Description:** A detailed description of the part, including usage and fitment notes.
- **Stock Level:** The number of items currently in stock. If the part is not precisely tracked in numbers, it will be marked as **Available/Not Available**.
- **Category:** Classification of the part based on its location (e.g., front, rear, inside, outside of the car).

#### 3.2 Part Addition Workflow

To maintain clarity and consistency in inventory, the system will prompt the user to provide the following information when adding a new part:

- **Brand:** Kia or Hyundai.
- **Model:** Specify the exact model, such as Kia Sorento or Hyundai Tucson.
- **Year/Year Range:** The year or a range of years the part fits (e.g., 2010-2015). A toggle option will be available to select either a single year or a range.
- **Location:** Define the part's location on the car (front, rear, inside, outside, or specific parts like "right door").

The user will also be able to add **custom descriptions** for each part to enhance future searches.

#### 3.3 Search and Filter System

To enhance ease of use, the system will include a powerful search and filter feature:

- **Search by Part Name:** Users can quickly locate parts by entering the part name (e.g., "Hood").

- **Search by Model, Year, and Brand:** Users can narrow down parts by selecting the car's brand, model, and year (with the option for single year or year ranges).
- **Search by Description:** Users can search using keywords from the part's description.
- **Filtering:** Results can be further filtered by stock availability (Available/Not Available) or location within the vehicle.

### 3.4 Availability Tracking

In cases where the exact stock count is not necessary, the system will allow users to mark parts as either **Available** or **Not Available**. This ensures that the shop can manage inventory fluidly without having to track every single item for low-demand parts.

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## 4. Database Architecture

The system will be based on a **relational database model**, ensuring data integrity and ease of scalability as the business grows. Key tables include:

### 4.1 Parts Table

This table will store core data about each part:

- **Part Name**
- **Serial Code**
- **Price**
- **Description**
- **Stock Level**
- **Availability Status** (Available/Not Available)
- **Category** (e.g., front, rear, inside, outside, etc.)

## 4.2 Category Table

This table will organize parts by car type:

- **Car Brand:** Kia or Hyundai.
- **Model:** e.g., Kia Sorento, Hyundai Elantra.
- **Year/Year Range:** Either a specific year or a range of years (e.g., 2010-2015).
- **Part Location:** Indicates whether the part belongs to the front, rear, interior, etc.

## 4.3 Search and Filter Algorithms

This section ensures the system's search feature is robust:

- **Keyword Matching:** Search will pull from part names, descriptions, and other attributes.
  - **Filters:** Users can apply filters for further precision, including brand, model, availability, and part location.
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## 5. User Interface (UI) Design

The system's user interface will be designed with ease of use in mind. Key UI components include:

### 5.1 Part Addition Form

- A clear, step-by-step form to add new parts with fields for serial code, part name, price, description, brand, model, year (or year range), location, and availability.

### 5.2 Search Interface

- A simple, user-friendly search bar with advanced filter options. Users can enter a part name, serial code, or description and refine results by brand, model, year, availability, and location.

### 5.3 Stock Status Indicator

- A visual display indicating whether a part is available or out of stock, helping users and staff make informed decisions quickly.

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## 6. Advantages of the Proposed System

### 6.1 Efficiency

The database will automate the inventory tracking process, reducing the time and effort required to manage car parts manually. By improving organization and search functionality, shop owners can quickly find parts and track availability.

### 6.2 Flexibility

The system will allow users to manage parts with or without specific stock counts. It also offers a flexible **year range selection** feature, which is essential for parts that fit multiple models across different years.

### 6.3 Improved Customer Service

With the ability to locate parts quickly and accurately, the shop can provide better service to customers, reducing waiting times and ensuring that parts are always available when needed.

### 6.4 Scalability

As the business grows, the system can scale to accommodate more parts, brands, models, and users. Whether deployed locally (localhost) or as a standalone application, the system will be adaptable to future needs.